

Technology Stack (Architecture & Stack)

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Team ID: Msu207Team5 6ab24d09e6af97f73bc88d13

Project Name: Import Data Using Transform Maps in Sheets

Maximum Marks: 4 Marks

Technical Architecture

The proposed system is designed to import data from **Google Sheets or CSV files** into a target system using **Import Sets and Transform Maps**. The architecture contains different layers such as data source, file upload, import processing, transformation, validation, database storage, and monitoring.

Architecture Flow

User

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Google Sheets / CSV File

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File Upload / Import Set

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Staging / Import Table

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Transform Map

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Field Mapping & Transformation Rules

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Data Validation

↓

Target Table / Database

↓

Import Status & Error Logs

Architecture Components

1. **User Layer** – User prepares and uploads the data.
2. **Data Source Layer** – Google Sheets or CSV acts as the source.
3. **Import Layer** – Import Set receives the uploaded data.
4. **Staging Layer** – Temporary import table stores source records before transformation.
5. **Transformation Layer** – Transform Map maps source fields to target fields.
6. **Validation Layer** – Checks missing, invalid, or incorrect data.
7. **Database Layer** – Successfully transformed records are stored.
8. **Monitoring Layer** – Import status and errors can be checked by the administrator.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Interface through which the user uploads the source file and checks import results.	ServiceNow Web Interface, HTML, CSS, JavaScript
2	Application Logic-1	Handles the overall import process and controls the movement of data from the source to the target system.	ServiceNow Platform, JavaScript
3	Application Logic-2	Handles source data import and stores the uploaded records temporarily before transformation.	Import Sets
4	Application Logic-3	Maps source fields to target fields and applies transformation rules.	Transform Maps
5	Data Validation	Checks data for missing values, invalid formats, incorrect fields, and other errors.	ServiceNow Validation, Business Rules
6	Database	Stores imported and processed records in the target tables.	ServiceNow Database
7	Staging / Import Table	Temporarily stores the data received from the uploaded Sheet or CSV before transformation.	Import Set Table
8	File Storage	Stores or provides the source CSV/file used for importing data.	CSV File / Local or Cloud File Storage

9	Data Source	Provides the original data that needs to be imported into the target system.	Google Sheets / CSV
10	Error Management	Records failed imports and provides information about errors during transformation.	Import Logs / Error Logs
11	Duplicate Management	Helps identify existing records and prevents unnecessary duplicate entries.	Transform Map Coalesce / Validation Rules
12	Monitoring	Allows administrators to check import status, successful records, and failed records.	ServiceNow Import History / Logs
13	Authentication	Controls access to the application and ensures only authorized users can perform import activities.	ServiceNow Authentication / Access Controls
14	Authorization	Defines which users can upload files, configure mappings, or manage import processes.	Roles and ACLs
15	Infrastructure (Server / Cloud)	Provides the environment required to run the application and store the processed data.	ServiceNow Cloud Platform

Components Not Required

External API-1

Purpose: External APIs are not required because the project can import data directly from Google Sheets or CSV files.

Technology: Not Required

External API-2

Purpose: The project does not require external APIs such as Aadhaar, Weather API, or payment APIs.

Technology: Not Required

Machine Learning Model

Purpose: Machine Learning is not required because the project mainly performs data importing, mapping, transformation, and validation.

Technology: Not Required

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The project uses web technologies and JavaScript-based platform functionality for application-level processing and customization.	HTML, CSS, JavaScript
2	Security Implementations	User access is controlled through authentication, roles, permissions, and access control. Only authorized users should perform import and configuration activities.	ServiceNow ACL, Roles, Authentication
3	Scalable Architecture	The system can handle increasing amounts of source data by using automated import processing instead of manual record entry.	Import Sets, Transform Maps, Cloud Platform
4	Availability	The cloud-based platform allows users to access the import functionality whenever the service is available.	ServiceNow Cloud Platform
5	Performance	Automated importing and transformation reduce manual processing time and allow multiple records to be processed efficiently.	Import Sets, Transform Maps
6	Maintainability	Transform Maps and transformation rules can be modified when the source or target data requirements change.	Transform Maps, JavaScript
7	Data Integrity	Field mapping, validation, and transformation rules help ensure that data is stored in the correct target fields.	Transform Maps, Validation Rules

8	Error Handling	Failed records and transformation errors can be identified through import logs and error information.	Import Logs, Error Logs
9	Data Consistency	Standard transformation and validation rules help maintain consistent data formats across imported records.	Transform Maps, Business Rules
10	Security & Access Control	Different users can be given different permissions depending on their responsibilities.	Roles, ACLs
11	Monitoring	Administrators can monitor import activities and check whether records were successfully imported.	Import History, System Logs
12	Extensibility	New mappings, fields, and transformation rules can be added when future project requirements change.	Tr